Hybrid Authorship and Digital Provenance: A Technical Methodology for XMP Metadata Injection

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DOI: 10.5281/zenodo.19193152

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1. Introduction

The integration of Artificial Intelligence into creative workflows has fundamentally disrupted traditional paradigms of authorship and intellectual property. By 2026, the binary classification of content as either strictly "human-made" or "AI-generated" is no longer sufficient to describe the reality of professional production. The vast majority of high-value digital assets are now the result of a **hybrid workflow**, where human conceptualization, prompt engineering, and iterative refinement are augmented by AI generation capabilities.

However, current legal frameworks and metadata standards lack the nuance to capture this spectrum. Claiming pure human authorship for an AI-assisted asset is factually inaccurate and poses legal risks. Conversely, attributing the work entirely to AI strips the human creator of their intellectual property rights, ignoring the creative direction and conceptual framework that guided the generation process.

This document outlines a pioneering technical methodology developed within the **Anonymo AI Framework 24x24 v2.0** to solve this identity problem. By leveraging the Extensible Metadata Platform (XMP) standard, we establish a self-contained, cryptographically verifiable provenance layer that travels with the visual asset, explicitly declaring its hybrid nature.

2. The Provenance Challenge

The core technical challenge in hybrid authorship is establishing a standardized, machine-readable, and tamper-evident method to declare the relationship between the human creator and the AI tool. Traditional approaches rely on external documentation, such as detached JSON files or README documents in repositories. While useful for academic transparency, these methods fail in practice because the metadata is easily separated from the asset when it is downloaded, shared, or published on third-party platforms.

Emerging industry initiatives, such as the Coalition for Content Provenance and Authenticity (C2PA), are attempting to address this by formalizing content credentials at scale. However, these standards are still in the early stages of adoption and often require complex infrastructure to implement.

The Anonymo AI Framework proposes a lightweight, robust, and immediately deployable solution: embedding custom metadata directly into the binary structure of the image file using XMP.

3. The XMP Solution

The Extensible Metadata Platform (XMP), defined by ISO 16684-1, is a standard created by Adobe for processing and storing standardized and proprietary metadata. XMP allows RDF/XML formatted data to be embedded directly into the binary of file formats such as JPEG, PNG, and TIFF.

3.1. Why XMP?

The primary advantage of XMP is that it **travels with the file**. The image carries its own identity. If a user downloads a JPEG containing XMP metadata, the provenance information remains intact. Furthermore, XMP is natively supported by a wide range of professional software, including Adobe Bridge, Lightroom, and command-line utilities like ExifTool, making the metadata easily accessible and verifiable.

3.2. The Custom `anonymo:` Namespace

To capture the specific nuances of hybrid authorship, the Anonymo AI Framework defines a custom XMP namespace: `https://anonymodocs.com/ns/metadata/1.0/`, registered with the prefix `anonymo:`. This namespace introduces fields that are exclusive to the framework, allowing creators to declare every dimension of the hybrid workflow.

Field	Example Value	Meaning
`anonymo:OriginalAuthorship`	Human	Declares that the core intellectual property and creative direction belong to a human.
`anonymo:CreationType`	Human-AI Hybrid	Specifies the nature of the workflow used to produce the asset.
`anonymo:IntellectualFramework`	TCF/TFB	Identifies the governing conceptual framework under which the asset was created.
`anonymo:IntegrityHash`	`e0f5497c...`	A SHA-256 hash representing the cryptographic proof of the pure visual content.
`anonymo:RecordLink`	`https://doi.org/...`	A link to an immutable public record, such as a Zenodo DOI.
`anonymo:ORCID`	`0009-0009-5364-249X`	The unique identifier of the human author.

4. Cryptographic Integrity Workflow

A critical vulnerability in metadata injection is that modifying the file to add metadata inherently changes the file's binary structure, and consequently, its cryptographic hash. If a hash is calculated *after* the metadata is injected, any subsequent modification to the metadata (even benign additions by other software) will invalidate the hash, breaking the chain of trust.

To ensure real integrity, the Anonymo AI Framework employs a strict 4-step workflow where the hash is calculated *before* injection.

Step 1: Generation

The visual asset is created through the human-AI hybrid workflow and saved as a raw binary file (e.g., a standard JPEG).

Step 2: SHA-256 Hash Calculation

A SHA-256 cryptographic hash is calculated on the raw image binary *before* any custom metadata is injected. This hash represents the pure visual content of the asset.

Step 3: XMP Injection

An XMP block is constructed containing the standard Dublin Core fields (title, creator, rights) and the custom `anonymo:` namespace fields. Crucially, the SHA-256 hash calculated in Step 2 is included in the `anonymo:IntegrityHash` field. This XMP block is then injected into the image's APP1 marker.

Step 4: Publication

The final asset, now containing its own proof of existence, is published to a public repository (e.g., GitHub) to establish a timestamp, and/or minted with a DOI (e.g., via Zenodo) to create an immutable academic record.

5. Verification and Prior Art

Because the `anonymo:IntegrityHash` represents the visual content prior to metadata injection, verification requires a specific process: extracting the visual data stream, stripping the metadata markers, and recalculating the hash to compare against the embedded value. This ensures that the visual content has not been tampered with, even if the metadata has been appended to by downstream systems.

By documenting and publishing this methodology, complete with functional Python injection scripts and a public GitHub repository linked to a Zenodo DOI, the Anonymo AI Framework establishes a clear record of prior art. While the broader market moves

toward complex credentialing systems, this work provides a functional, verifiable standard for hybrid authorship today.

6. Conclusion

Hybrid authorship is the present reality of digital creation; verifiable provenance is its necessary future. The methodology developed within the Anonymo AI Framework solves a problem the industry is still trying to formulate. By turning every asset into its own proof of existence through XMP injection and cryptographic hashing, we ensure that human creators retain their intellectual property while transparently acknowledging the role of AI in the creative process.

For technical implementation details, scripts, and the complete framework architecture, refer to the official repository: i-hatr-tcf-tfb-integration.